A brief overview of Brazil's energy planning

INDIA SMART GRID FORUM

3rd INDIA – BRAZIL SMART ENERGY WORKSHOP

Thiago Ivanoski Teixeira

Division of Energy Economics and Socioenvironmental Studies

March 12th, 2026

Energy Research Office - EPE

State-owned company attached to the Brazilian Ministry of Mines and Energy (MME)

Development of integrated energy planning studies and energy statistics to support formulation, implementation and evaluation of the Brazilian energy policy

www.epe.gov.br

Permanent Member of Energy Policy National Council (CNPE)

International reference to understand EPE

U.S. DEPARTMENT OF ENERGY ⟷ eia U.S. Energy Information Administration

epe ⟷ epe

Some studies and reports of EPE

Integrated approach of the Energy sector, which includes**:** O&G, Natural Gas, Biofuels, Electricity, Nuclear, Energy efficiency, Socialenvironmental issues...

Brazil in big numbers – rich in energy resources

Brazil is the **largest economy** in **Latin America**

215 million people

7th largest population in the world

GDP of **$2.2 trillion**

10th largest economy in the world

Pre-salt has boosted oil and gas reserves and Brazil is on track to become one of top producers

4th largest agricultural production in the world represents imense bioenergy potential

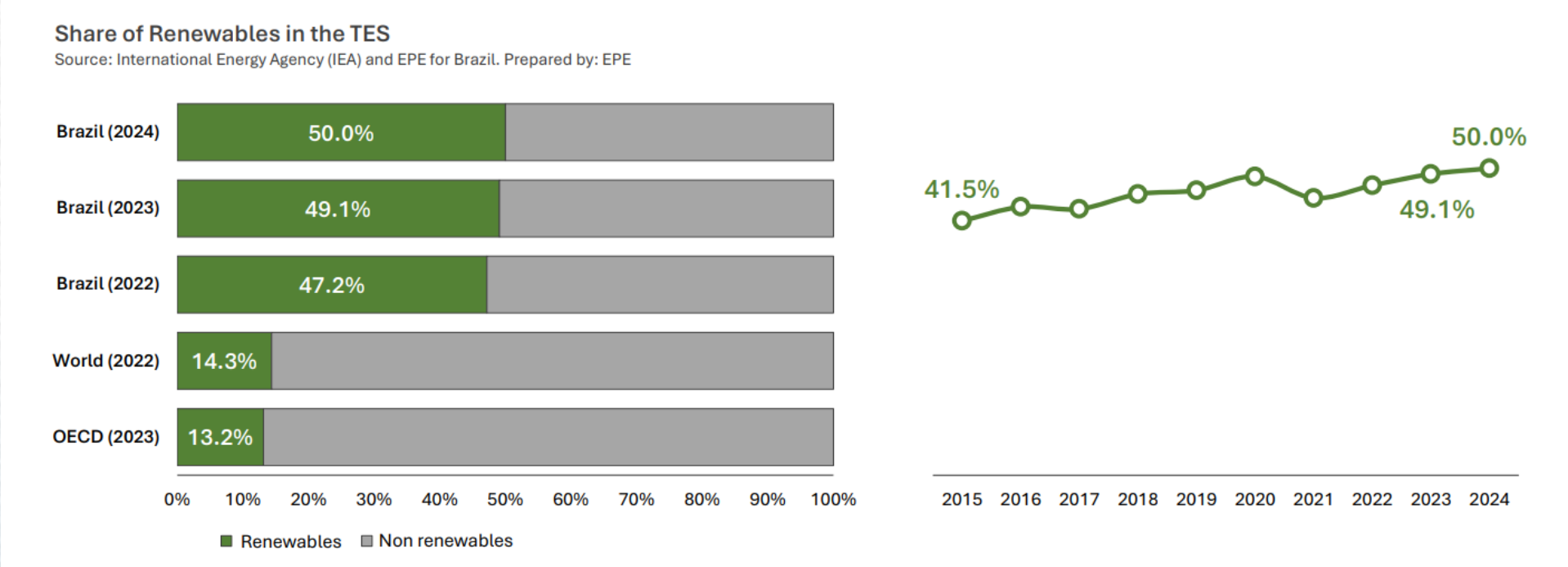
Huge and high quality onshore windpower potential and yet untapped offshore wind

Excellent irradiation covers the territory and best PV sites alone would exceed electricity demand

Uranium reserves enough to supply at least 10 additional 1 GW Nuclear Power Plants

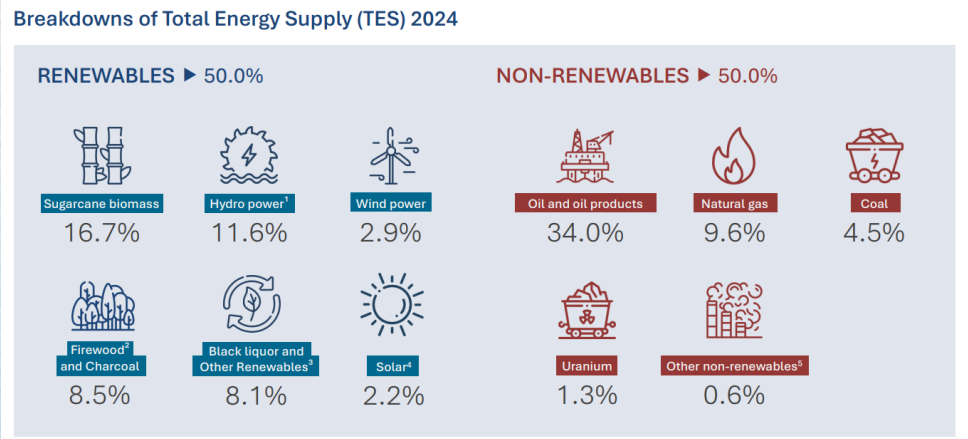
World's 2nd largest hydropower producer. Pumped storage potential is huge as well

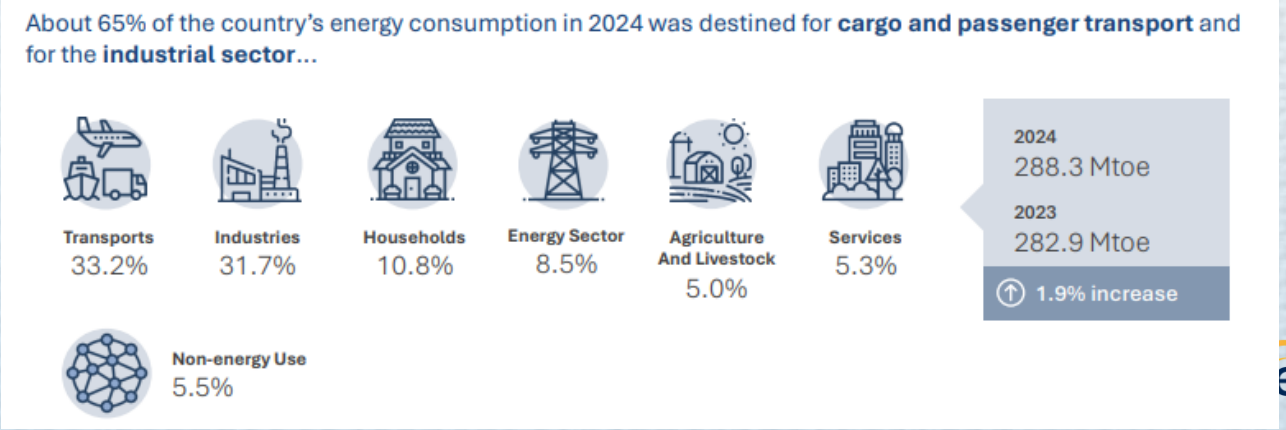
Overview of Brazil's Energy Matrix



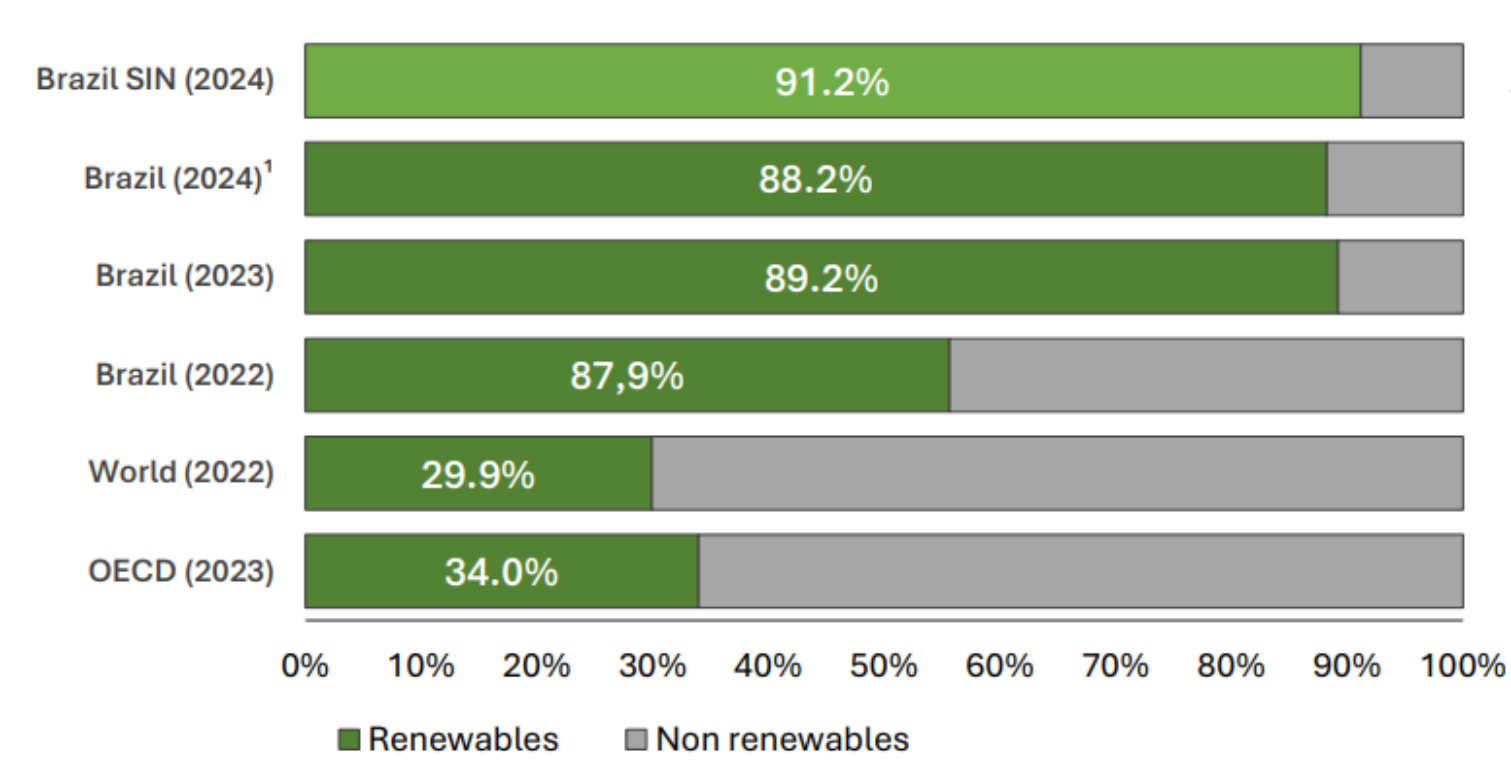
Brazilian Energy Balance

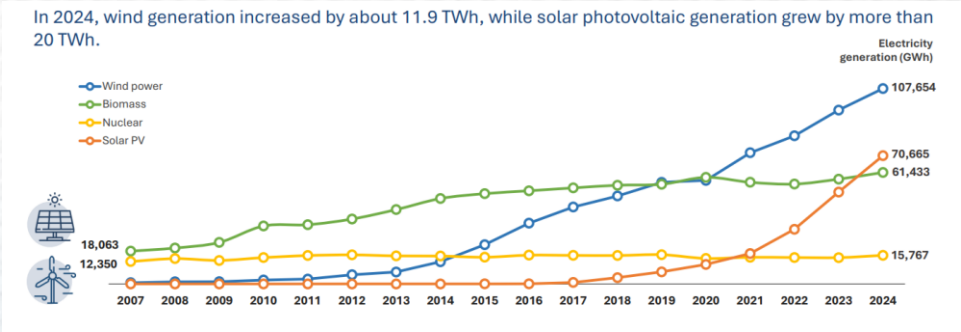
https://www.epe.gov.br/sites-pt/publicacoes-dados-abertos/publicacoes/PublicacoesArquivos/publicacao-885/topico-767/BEN_S%C3%ADntese_2025_EN.pdf

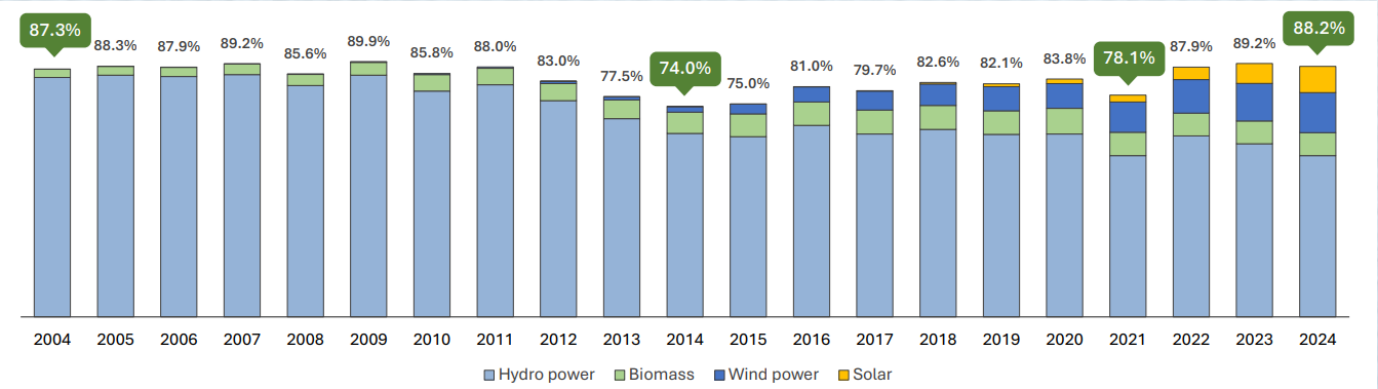




Overview of Brazil's Electricity Matrix







Brazilian Energy Balance

https://www.epe.gov.br/sites-pt/publicacoes-dados-abertos/publicacoes/PublicacoesArquivos/publicacao-885/topico-767/BEN_S%C3%ADntese_2025_EN.pdf

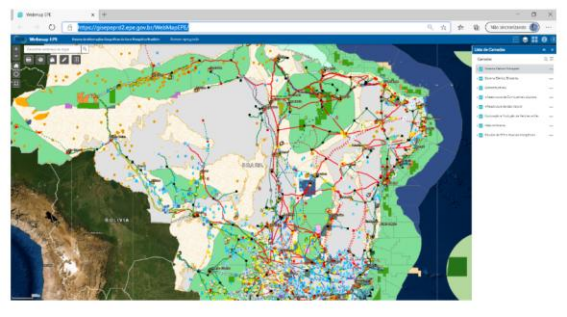
Brazilian Power System

- Area: 8,516,000 km^2
- Installed generation capacity: 215 GW (excl. DG). ~ 50% hydro
- HV transmission system length: ~180,000 km (~99% of the load is connected) -> and expanding… (Auctioned more than US$ 12 bi CAPEX in the last 3 years)
- Growing electricity and investment market
- Private participation in G, T, D and trading (nuclear: state-owned by constitution)

Existing Power Grid (HV)

Off-grid Systems

EPE´s WEBMAP tool ->



https://www.epe.gov.br/en/publications/publications/webmap-epe

Energy Planning: What Are Our Compasses?

National Energy Plan (PNE)

Long-term strategic instrument

- **Time Horizon: 30 years (revised every 5 years)**
- Guides policies, programs, and initiatives
- Focus: sustainability, security, affordability
- Basis for the national energy strategy

Ten-Year Energy Expansion Plan (PDE)

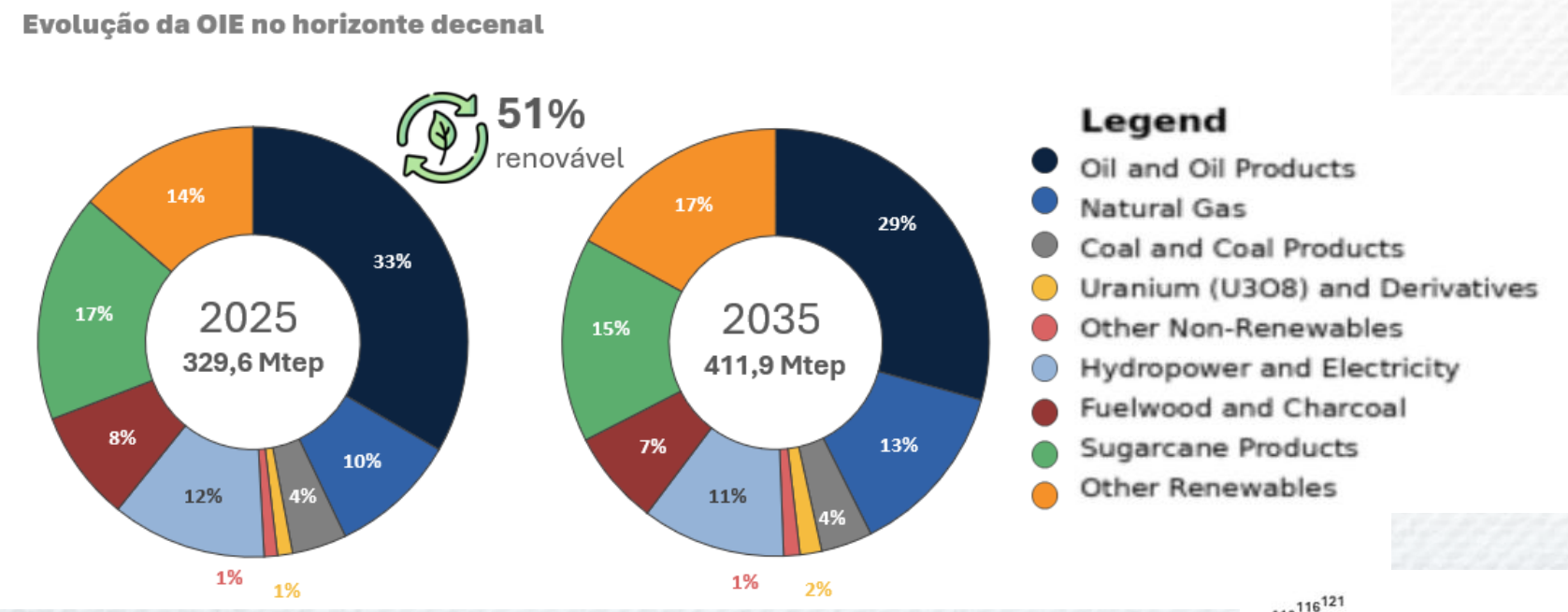
Medium-term technical planning

- **Time Horizon: 10 years (revised annually)**
- Details supply and infrastructure expansion
- Reflects current policies, market signals, and technical feasibility
- Basis for investment and regulatory decisions

PNE 2055 and PDE 2035 in Public Consultation Process until March, 14th

PDE 2035 – Ten-year Energy Plan

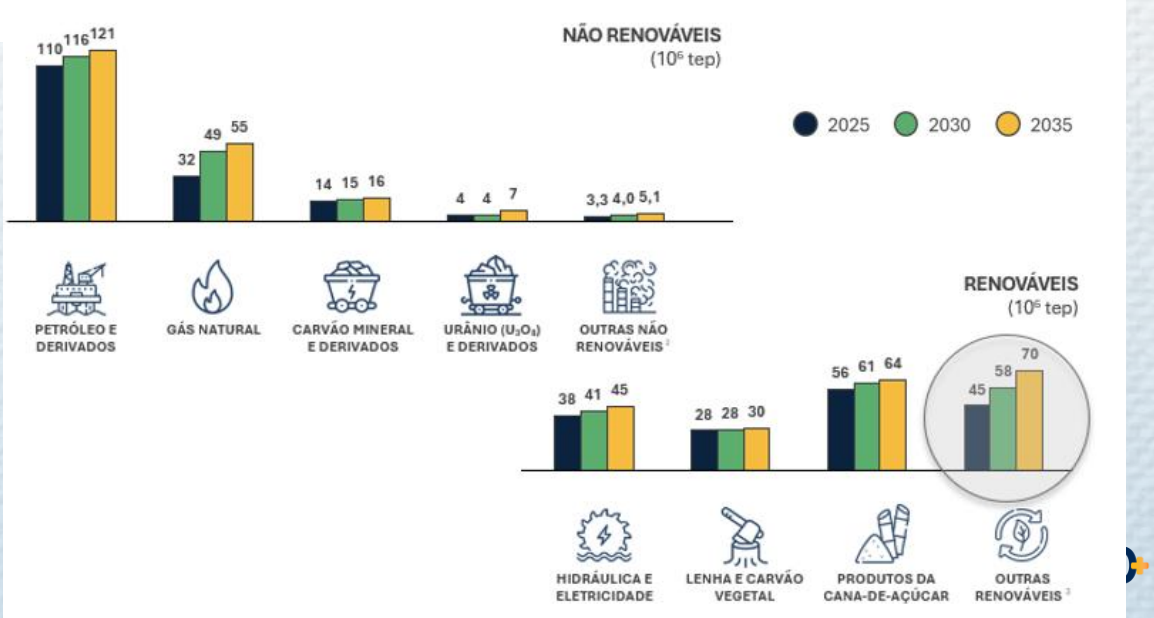
PNE 2055 and PDE 2035 in Public Consultation Process until March, 14th



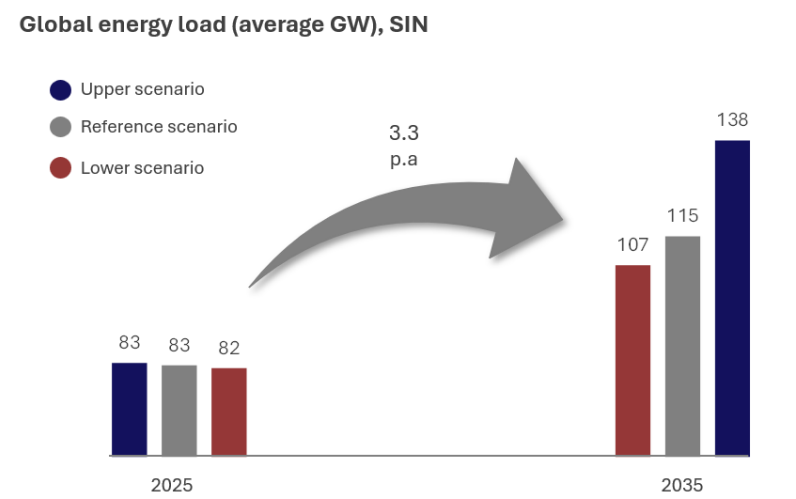
ENERGIA ELÉTRICA R$ 596 Bilhões (17%)	PETRÓLEO E GÁS NATURAL R$ 2.818 Bilhões (80%)	BIOCOMBUSTÍVEIS LÍQUIDOS R$ 115 Bilhões (3%)
GERAÇÃO CENTRALIZADA R$ 374 Bilhões (11%)	E&P DE PETRÓLEO E GÁS NATURAL R$ 2.611 Bilhões (74%)	ETANOL[5] R$ 72 Bilhões (2%)
GERAÇÃO DISTRIBUÍDA (MICRO E MINIGERAÇÃO) R$ 106 Bilhões (3%)	OFERTA DE DERIVADOS DE PETRÓLEO R$ 157 Bilhões (4%)	BIODIESEL / BIOMETANO/BECCS R$ 15 Bilhões (0,4%)
TRANSMISSÃO[4] R$ 117 Bilhões (3%)	OFERTA DE GÁS NATURAL R$ 50 Bilhões (1%)	SAF/ DIESEL VERDE R$ 28 Bilhões (0,8%)

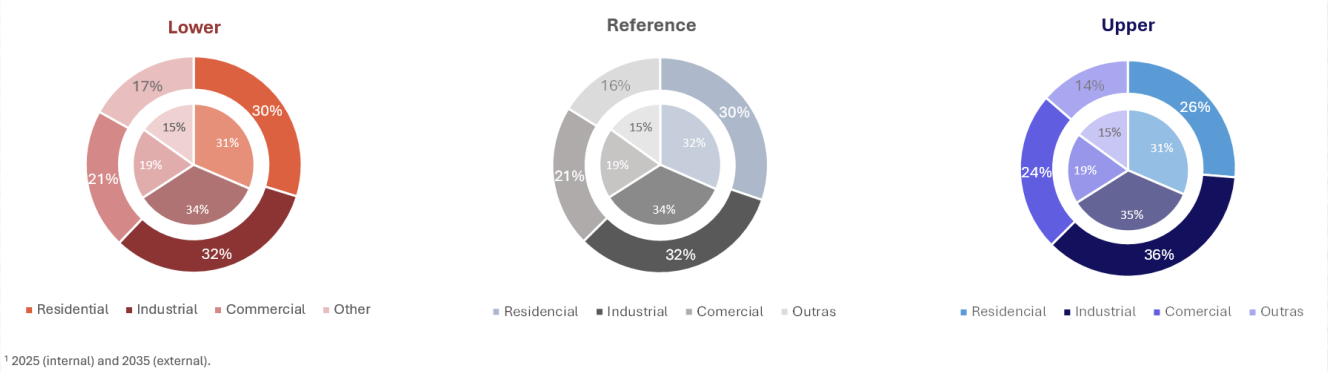
[4] Inclui instalações já licitadas que entrarão em operação no período decenal; [5] Inclui investimentos para formação de canaviais e unidades de etanol 1G, 2G e de cereais. Não inclui açúcar.

Taxa de câmbio referencial: R$ 6,10 / US$ (dez/2024).



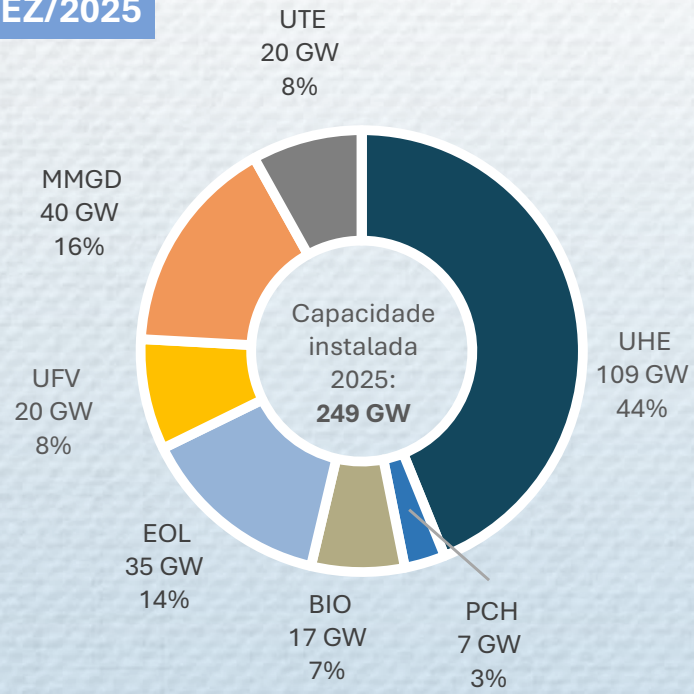
PDE 2035 – Ten-year Energy Plan



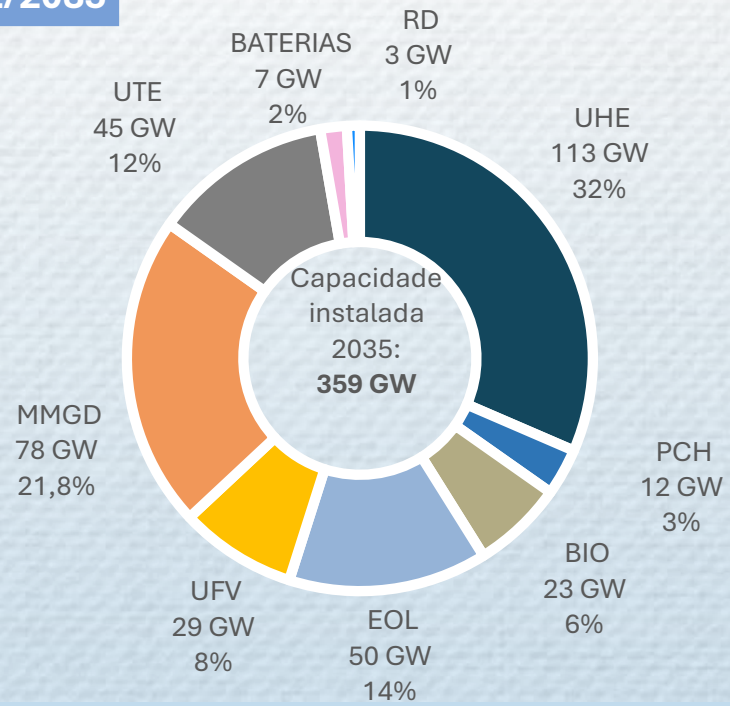


[1] 2025 (internal) and 2035 (external).

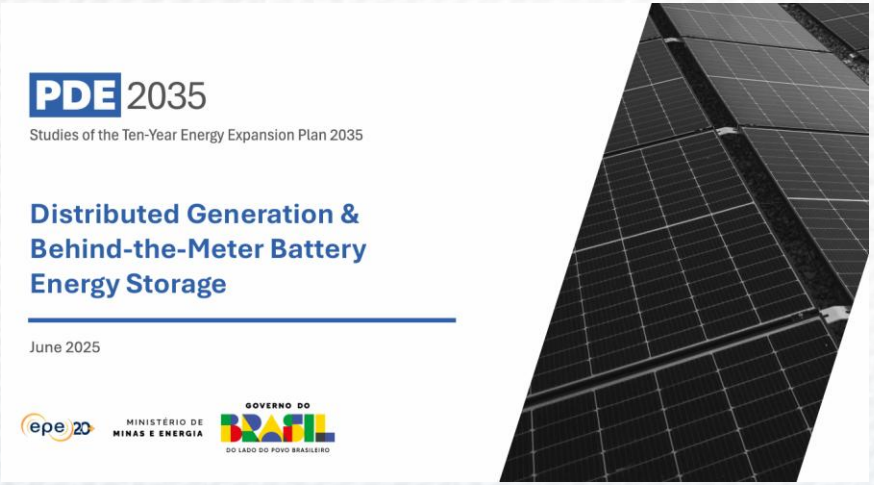
DEZ/2025

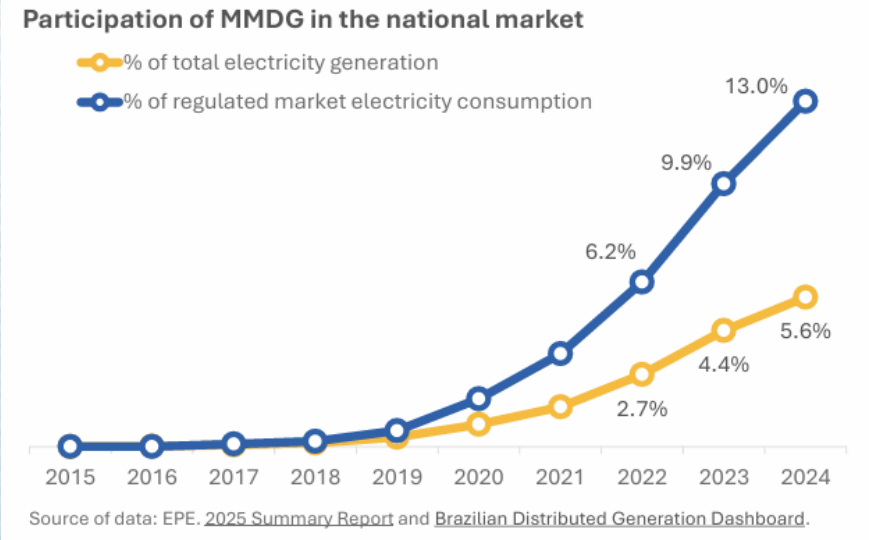


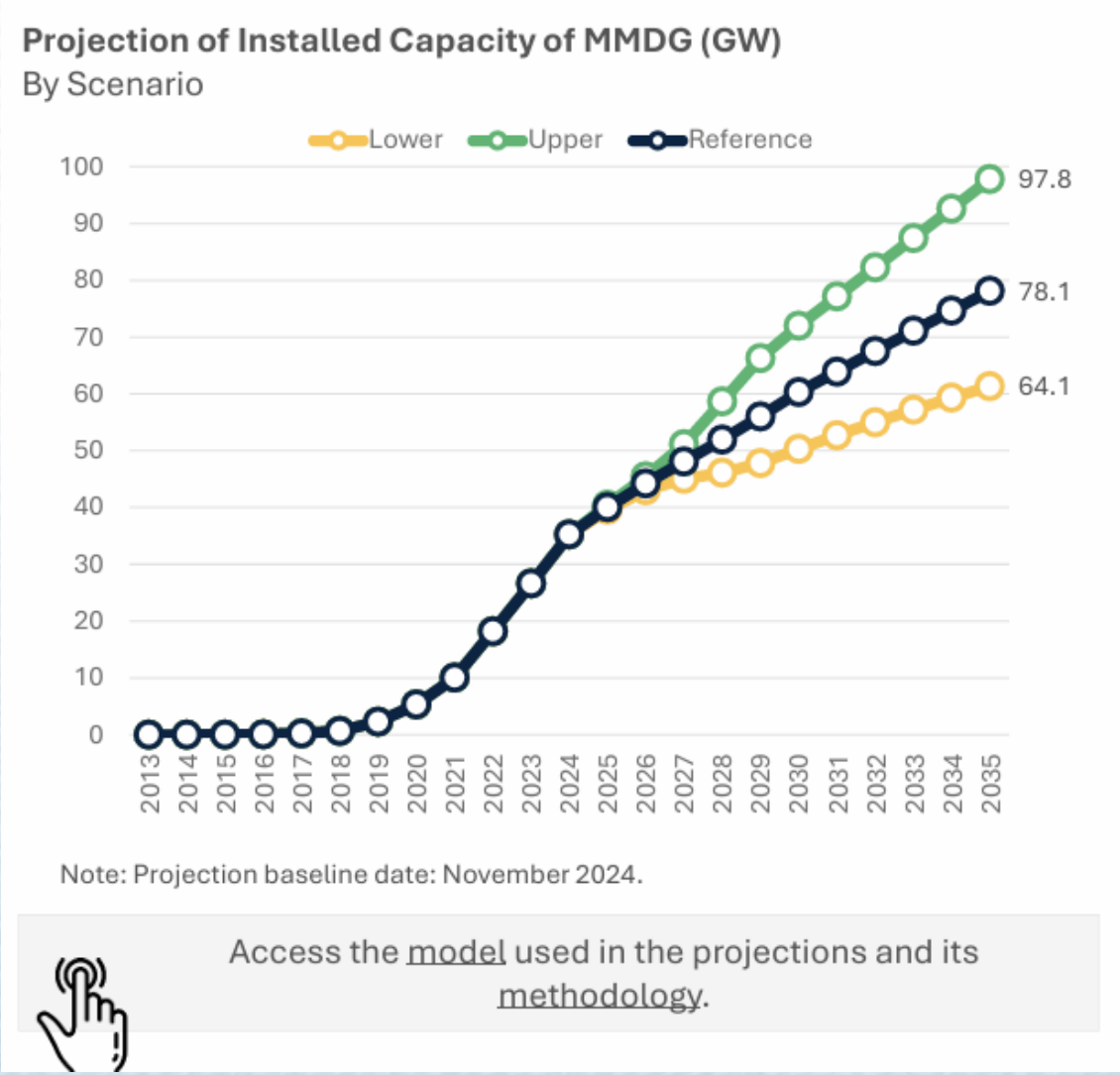
DEZ/2035



PDE 2035 – Distributed Generation & Behind-the-Meter Battery Energy Storage







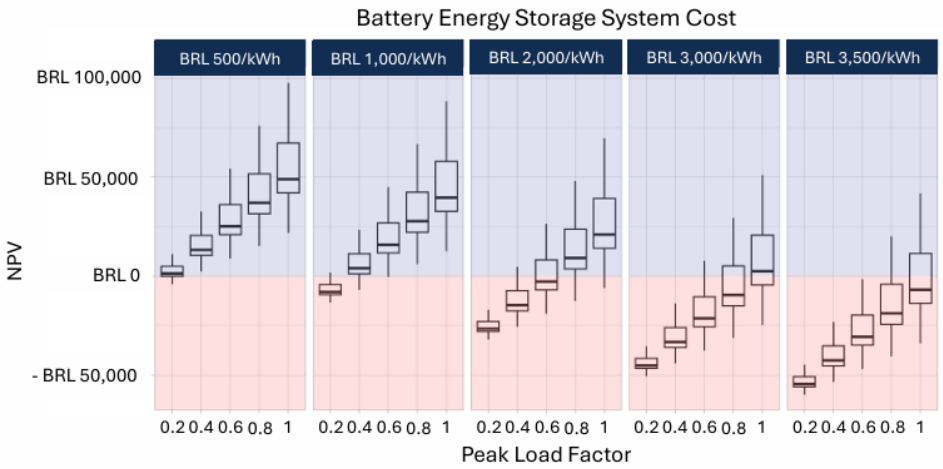
Link

PDE 2035 – Distributed Generation & Behind-the-Meter Battery Energy Storage

	Financial support	Dynamic tariffs	Electricity market access
Germany	High	High	Medium
Australia	High	High	High
Canada	Medium	Medium	Medium
China	Medium	High	Medium
South Korea	Low	Medium	Medium
USA	High	High	Medium
India	Low	Medium	Low
Italy	High	High	High
Japan	High	High	Medium
United Kingdom	High	High	High
Brazil	Medium	Medium	Low

Support Level: Low, Medium, High

Source: Adapted from IEA (2024)
[1] Conversion A$1 AUD = BRL 3.615 (commercial exchange rate at 11:30 on 06/26/2025)

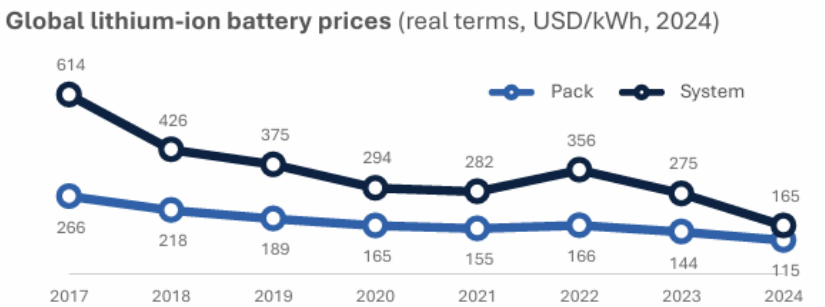


Understanding the Peak Load Factor (PLF)

The annual Peak Load Factor (PLF) expresses the ratio between the average consumption (MW average) and the maximum demand of a user during the peak period (MW).

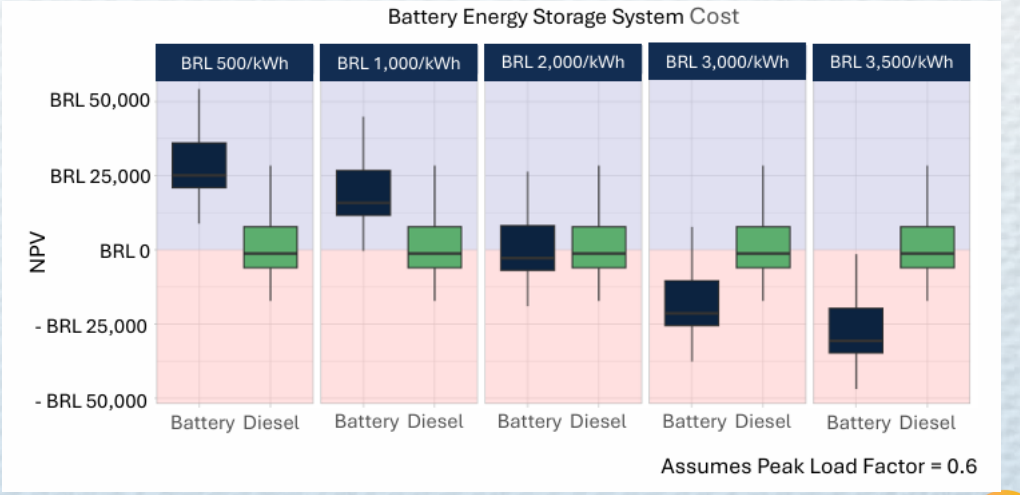
Low PLF values indicate low utilization of the BESS, increasing the payback period of the investment, whereas high PLF values reflect greater stability of the peak consumption profile and, therefore, greater economic feasibility of storage.

- Results show that for consumers with a high PLF, BESS installation may already be economically viable under current tariff conditions in some distribution utilities.



Note: The system price accounts for all project equipment, excluding EPC and grid interconnection.
Source: BloombergNEF (2024; 2025)

Link



PDE 2035 – Distributed Generation & Behind-the-Meter Battery Energy Storage

New applications for storage systems

As the regulation of energy storage systems advances, new solutions can be envisioned to ensure the efficiency and security of supply in a context of increasing decentralization.

Aggregators and Virtual Power Plants (VPP)

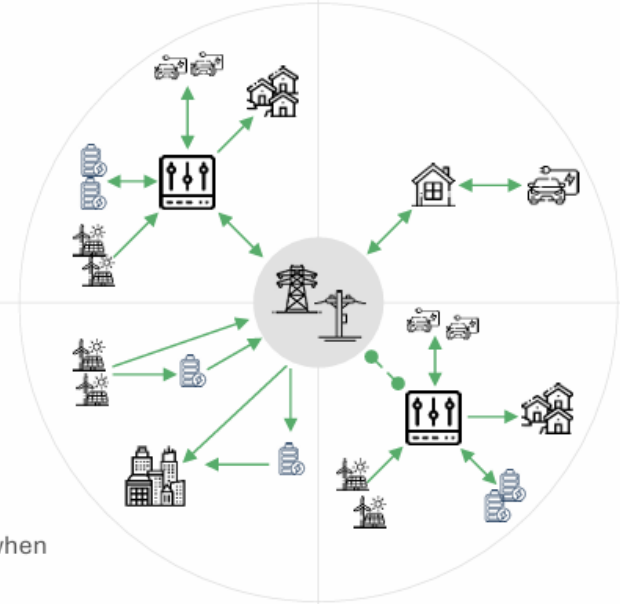
Aggregators use centralized systems that **control Distributed Energy Resources (DERs) as a single system (VPP)**, optimizing their operation.

- Local flexibility for distributors
- Balancing services for transmission operators
- Shifting of larger loads

Virtual Power Lines

Large grid-connected BESS to meet demand and supply.

- On the supply side, they store generation surpluses when there are grid constraints
- On the demand side, they charge when grid capacity allows and unload when needed



Vehicle-to-Grid (V2G)

Allows **electricity to flow from the grid to the electric vehicle and from the vehicle to the grid**.

- Smoothing load and generation variations
- Reduction of vehicle total cost of ownership
- Consumer load balancing
- Individual consumer flexibility

Microgrids

Grids **capable of disconnecting from the main system and running independently** for a period.

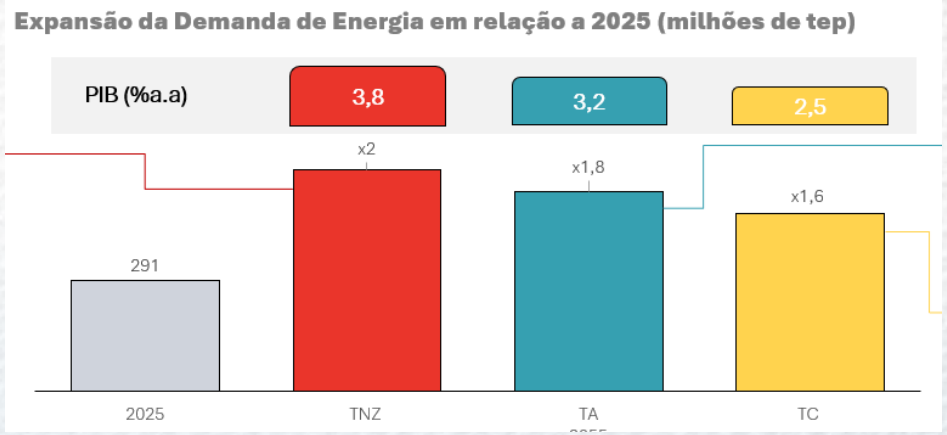
- Increased reliability within the microgrid
- Improved power quality within the microgrid
- Renewables within the microgrid operate even with unavailability of the main grid

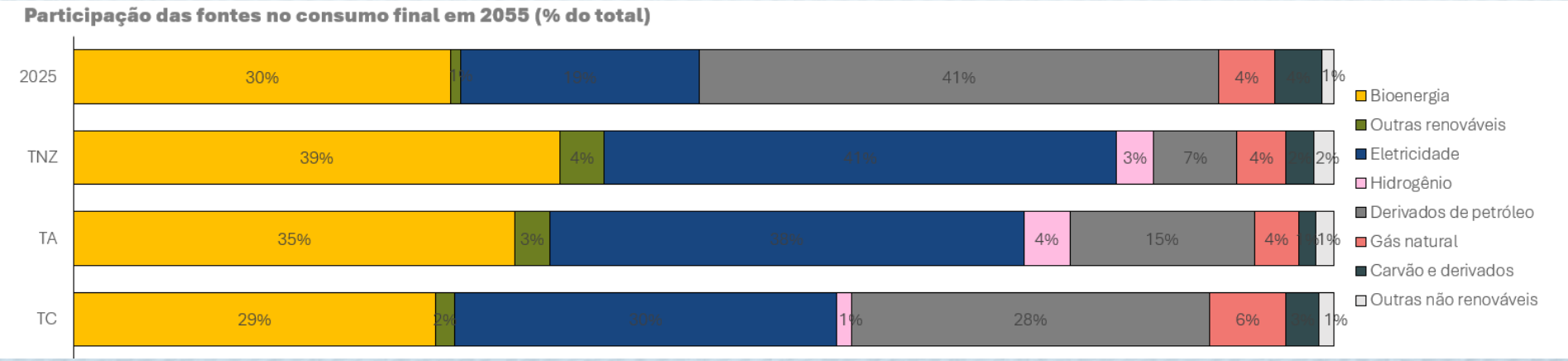
Source: Adapted from IRENA (2019, 2020)

Link

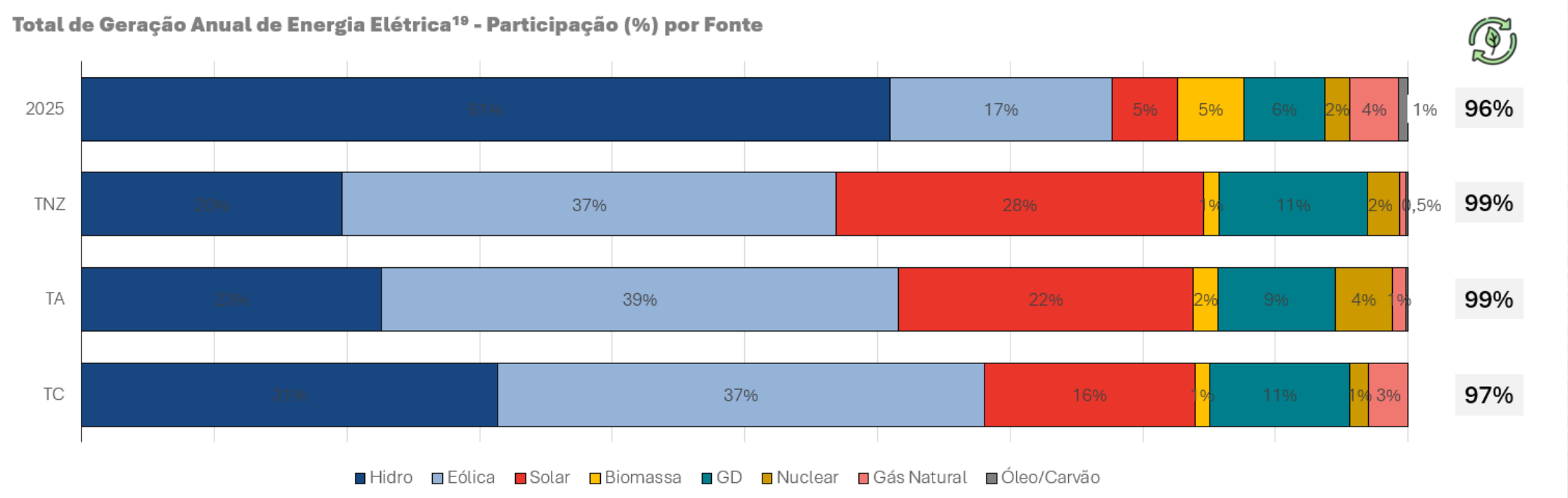
PDE 2055 – National Energy Plan

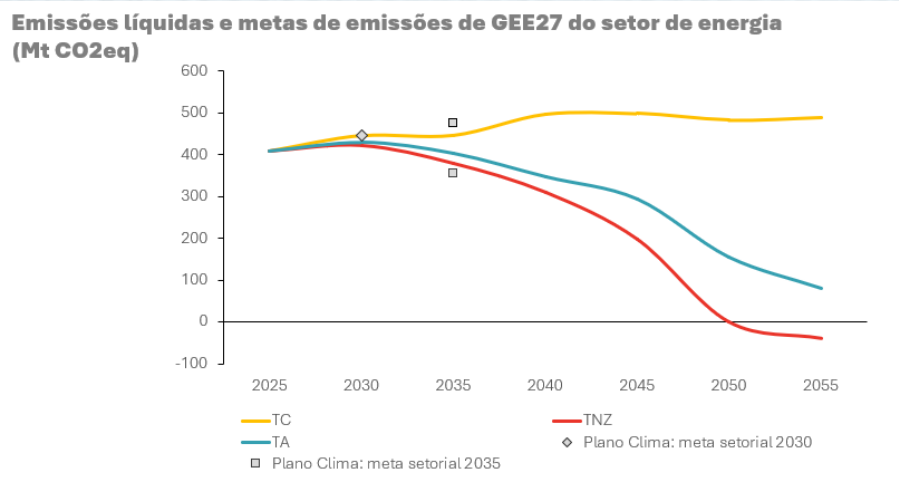
PNE 2055 and PDE 2035 in Public Consultation Process until March, 14th

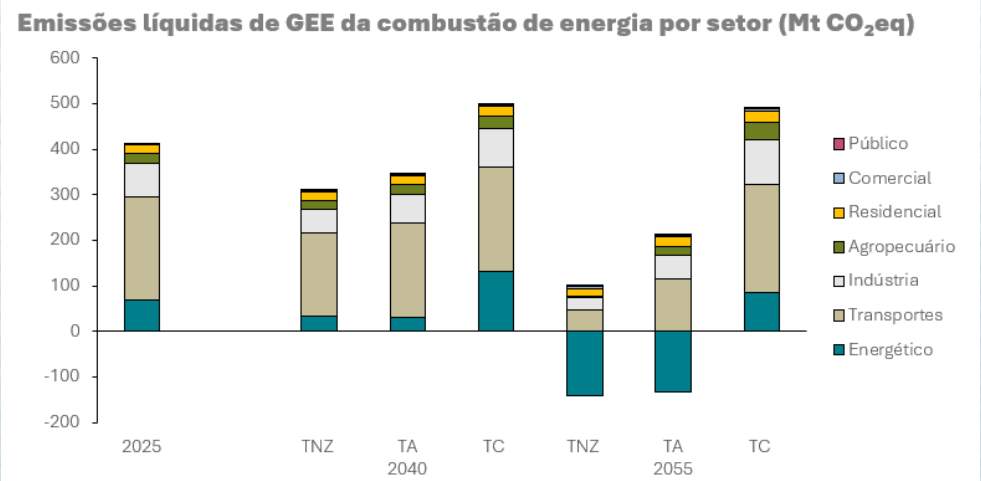




PDE 2055 – National Energy Plan







The Challenge of Climate Change

Fonte: Atlas do IPCC (2021)

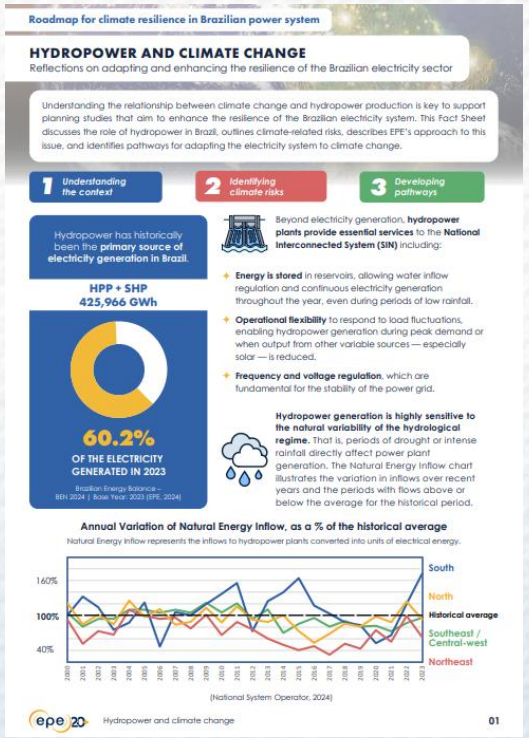
Projections for the next 20 to 30 years are consistent with a global warming of 2°C compared with a similar period between 1960–2014

- Increase in the average air temperature throughout the country
- Increase in the intensity and frequency of temperature extremes (both heat and cold) and heatwaves.
- Increase in average precipitation in the SES region and reduction in the NES and NSA regions
- Increase in the intensity and frequency of heavy rainfall and flooding across the 4 regions, with emphasis on SES and the Amazon.
- Growing trends in the frequency and duration of droughts in the SAM, NSA, and NES regions
- Increase in surface radiation in the NSA, SAM, and NES regions
- Aumento da radiação na superfície nas regiões NSA, SAM e NES

Some recente studies of Climate Change – Energy planning perspective

https://www.epe.gov.br/pt/publicacoes-dados-abertos/publicacoes/estudo-de-impactos-das-mudancas-climaticas-no-planejamento-da-geracao-de-energia-eletrica

https://www.epe.gov.br/sites-pt/publicacoes-dados-abertos/publicacoes/Paginas/Roadmap-para-o-Fortalecimento-da-Resiliencia-do-Setor-Eletrico-em-Resposta-as-Mudancas-Climaticas-Revisao-Bibliografica.aspx



https://www.epe.gov.br/pt/publicacoes-dados-abertos/publicacoes/roadmap-para-o-fortalecimento-da-resiliencia-do-setor-eletrico-em-resposta-as-mudancas-climaticas

The Role of Energy Planning in the Global Transition

G20 BRASIL 2024
BUILDING A JUST WORLD AND A SUSTAINABLE PLANET

PRINCIPLES FOR JUST AND INCLUSIVE ENERGY TRANSITIONS

FOZ DO IGUAÇU, 04 OCTOBER 2024

1 – Energy planning for just and inclusive energy transitions

Acknowledge the importance of long-term regional and domestic energy planning and policies across various sectors to guide actions and financing mechanisms that promote energy transitions and design and implement just and inclusive energy transition policies in individual countries, while ensuring energy security, affordability, accessibility, and markets stability and economic prosperity.

- Energy planning has regained relevance as a tool to:
  - Mobilize investment
  - Strengthen energy security
  - Reduce risks for infrastructure projects
  - Tackle energy poverty

Strategic functions to foster a favorable investment environment

A multi-tiered approach: linking short-term actions to long-term vision

From planning to projects: bridging the gap between vision and implementation

Addressing energy poverty through inclusive planning

The role of strong institutions in building trust and stability

GLOBAL COALITION FOR ENERGY PLANNING

epe 20+

Conclusions

- ✓ **Brazil:** Strategic global position, clean energy matrix, natural resources, geographic location, major consumer market, and great potential. **World leader in the Energy Transition**/Transformation.
- ✓ **The energy transition** is not a turn of the key, **it is a process!** **Innovation and Digitalization** are very important issues!
- ✓ A decade of change: **market designs and redesigns**, developments and transformations in segments of the energy sector, technological improvements, innovation and new business models.
- ✓ **Energy Planning: constant adaptation and improvement of studies,** ensuring energy security.
- ✓ Seizing opportunities for socioeconomic development in pursuit of a **just and inclusive energy transition.**

Thank you!

www.epe.gov.br

Praça Pio X, 54 – Centro, Rio de Janeiro/RJ